

Exercise – Understanding simultaneity bias using simulation

1. Aim of the exercise

The aim of this exercise is to understand simultaneity bias through simulation.

2. Theory

Consider the simultaneous equations model

$$\begin{aligned}y_{1i} &= \alpha_1 y_{2i} + \beta_1 z_{1i} + u_{1i}, \\ y_{2i} &= \alpha_2 y_{1i} + \beta_2 z_{2i} + u_{2i}.\end{aligned}$$

For simplicity, the intercepts are omitted.

Assume that

$$\begin{aligned}\mathbb{E}[u_{1i} \mid z_{1i}, z_{2i}] &= 0, \\ \mathbb{E}[u_{2i} \mid z_{1i}, z_{2i}] &= 0.\end{aligned}$$

By the law of iterated expectations, these assumptions imply

$$\begin{aligned}\mathbb{E}[u_{1i}] &= 0, \\ \mathbb{E}[u_{2i}] &= 0.\end{aligned}$$

They also imply

$$\mathbb{E}[z_{ji}u_{ki}] = 0, \quad j, k \in \{1, 2\},$$

so that both exogenous variables, z_{1i} and z_{2i} , are uncorrelated with both disturbance terms, u_{1i} and u_{2i} .

Suppose that the parameter of interest is α_1 in the first equation,

$$y_{1i} = \alpha_1 y_{2i} + \beta_1 z_{1i} + u_{1i}.$$

To determine whether α_1 can be estimated consistently by OLS, we must examine whether the regressor y_{2i} is uncorrelated with the disturbance term u_{1i} .

To determine whether y_{2i} is correlated with u_{1i} , we solve the system for y_{2i} in terms of the exogenous variables and the disturbance terms.

Substituting the first equation into the second equation gives

$$\begin{aligned}y_{2i} &= \alpha_2 y_{1i} + \beta_2 z_{2i} + u_{2i} \\ &= \alpha_2 (\alpha_1 y_{2i} + \beta_1 z_{1i} + u_{1i}) + \beta_2 z_{2i} + u_{2i} \\ &= \alpha_1 \alpha_2 y_{2i} + \beta_1 \alpha_2 z_{1i} + \alpha_2 u_{1i} + \beta_2 z_{2i} + u_{2i}.\end{aligned}$$

Collecting the terms involving y_{2i} yields

$$(1 - \alpha_1 \alpha_2) y_{2i} = \beta_1 \alpha_2 z_{1i} + \beta_2 z_{2i} + \alpha_2 u_{1i} + u_{2i}.$$

Assuming that

$$1 - \alpha_1 \alpha_2 \neq 0,$$

we obtain the reduced-form expression

$$y_{2i} = z_{1i} \frac{\beta_1 \alpha_2}{1 - \alpha_1 \alpha_2} + z_{2i} \frac{\beta_2}{1 - \alpha_1 \alpha_2} + u_{1i} \frac{\alpha_2}{1 - \alpha_1 \alpha_2} + u_{2i} \frac{1}{1 - \alpha_1 \alpha_2}.$$

The parameter of interest is α_1 in the first equation,

$$y_{1i} = \alpha_1 y_{2i} + \beta_1 z_{1i} + u_{1i}.$$

To estimate α_1 consistently by OLS, the regressor y_{2i} must be uncorrelated with the disturbance term u_{1i} , that is,

$$\mathbb{E}[y_{2i} u_{1i}] = 0.$$

Does this condition hold?

The reduced-form expression for y_{2i} contains u_{1i} whenever $\alpha_2 \neq 0$. This suggests that y_{2i} may be correlated with the disturbance term in the first equation. To verify this formally, we compute $\mathbb{E}[y_{2i} u_{1i}]$.

Consider again the reduced-form expression

$$y_{2i} = z_{1i} \frac{\beta_1 \alpha_2}{1 - \alpha_1 \alpha_2} + z_{2i} \frac{\beta_2}{1 - \alpha_1 \alpha_2} + u_{1i} \frac{\alpha_2}{1 - \alpha_1 \alpha_2} + u_{2i} \frac{1}{1 - \alpha_1 \alpha_2}.$$

Multiplying both sides by u_{1i} and taking expectations yields

$$\begin{aligned} \mathbb{E}[y_{2i} u_{1i}] &= \mathbb{E}[z_{1i} u_{1i}] \frac{\beta_1 \alpha_2}{1 - \alpha_1 \alpha_2} + \mathbb{E}[z_{2i} u_{1i}] \frac{\beta_2}{1 - \alpha_1 \alpha_2} \\ &\quad + \mathbb{E}[u_{1i}^2] \frac{\alpha_2}{1 - \alpha_1 \alpha_2} + \mathbb{E}[u_{2i} u_{1i}] \frac{1}{1 - \alpha_1 \alpha_2}. \end{aligned}$$

Since

$$\mathbb{E}[z_{1i} u_{1i}] = \mathbb{E}[z_{2i} u_{1i}] = 0,$$

it follows that

$$\mathbb{E}[y_{2i} u_{1i}] = \mathbb{E}[u_{1i}^2] \frac{\alpha_2}{1 - \alpha_1 \alpha_2} + \mathbb{E}[u_{2i} u_{1i}] \frac{1}{1 - \alpha_1 \alpha_2}.$$

This shows that $\mathbb{E}[y_{2i} u_{1i}] \neq 0$ whenever either

$$\alpha_2 \neq 0,$$

or

$$\mathbb{E}[u_{2i} u_{1i}] \neq 0.$$

Consider again the simultaneous equations model

$$\begin{aligned} y_{1i} &= \alpha_1 y_{2i} + \beta_1 z_{1i} + u_{1i}, \\ y_{2i} &= \alpha_2 y_{1i} + \beta_2 z_{2i} + u_{2i}. \end{aligned}$$

We have shown that

$$\mathbb{E}[y_{2i} u_{1i}] \neq 0.$$

Therefore, in the first equation, the regressor y_{2i} is correlated with the disturbance term u_{1i} . The exogeneity assumption required for OLS is violated and, consequently, the OLS estimator of α_1 is generally biased and inconsistent. This phenomenon is known as simultaneity bias.

3. Application

This application simulates data from a simultaneous equations model to illustrate how simultaneity generates endogeneity. A sample of 1,000 observations is generated using prespecified structural parameters, exogenous variables, and structural disturbances. The endogenous variables are then obtained from the reduced-form solution of the system.

To demonstrate the violation of the exogeneity assumption, the sample correlation between y_2 and the structural disturbance u_1 is computed. Since the reduced-form expression for y_2 contains a component of u_1 , the correlation is expected to be nonzero. This provides a numerical illustration of the fact that an endogenous regressor is correlated with the error term, which leads to inconsistent ordinary least squares estimation in simultaneous equations models.

```
8 %% 3. Application
9
10 % 3.1. Clear memory
11 clear;
12
13 % 3.2. Set sample size
14 N_obs = 1000;
15
16 % 3.3. Set structural parameters
17 alpha_1 = 0.5;
18 alpha_2 = 0.6;
19 beta_1 = 1.0;
20 beta_2 = 1.0;
21
22 % 3.4. Generate exogenous variables
23 z_1 = random('Normal',0,1,[N_obs 1]);
24 z_2 = random('Normal',0,1,[N_obs 1]);
25
26 % 3.5. Generate structural errors
27 u_1 = random('Normal',0,1,[N_obs 1]);
28 u_2 = random('Normal',0,1,[N_obs 1]);
29
30 % 3.6. Solve the simultaneous system
31 y_1 = (beta_1 * z_1 + alpha_1 * beta_2 * z_2 + ...
32       u_1 + alpha_1 * u_2) / ...
33       (1 - alpha_1 * alpha_2);
34
35 y_2 = (alpha_2 * beta_1 * z_1 + beta_2 * z_2 + ...
36       alpha_2 * u_1 + u_2) / ...
37       (1 - alpha_1 * alpha_2);
38
39 % 3.7. Compute the correlation between y_2 and u_1
40 corr_y2_u1 = corr(y_2,u_1);
```

4. OLS estimation of the first structural equation

The first structural equation is estimated using ordinary least squares (OLS). The regressor

matrix is constructed from the endogenous variable y_2 and the exogenous variable z_1 , corresponding to the specification of the first structural equation.

OLS estimation is then applied to obtain the coefficient estimates. Particular attention is given to the estimate of α_1 , which measures the effect of y_2 on y_1 . Since y_2 is correlated with the structural disturbance u_1 , the OLS estimate is expected to be biased and inconsistent. This illustrates the simultaneity bias that arises when an endogenous variable is treated as an exogenous regressor.

```

8  %% 4. OLS estimation of the first structural equation
9
10 % 4.1. Create the systematic part of the regression
11 X = [y_2 z_1];
12
13 % 4.2. Obtain OLS statistics
14 LSS = lss(y_1,X);
15
16 % 4.3. Obtain the OLS estimate of alpha_1
17 alpha_1_hat = LSS.B_hat(1,1);

```

5. Sampling distribution of the OLS estimator

To investigate the finite-sample properties of the OLS estimator, a Monte Carlo simulation with 1,000 replications is conducted. For each replication, new realizations of the exogenous variables and structural disturbances are generated, and the simultaneous equations system is solved to obtain the endogenous variables.

The first structural equation is then estimated by OLS, and the resulting estimate of α_1 is stored. Repeating this procedure many times produces an empirical sampling distribution for $\hat{\alpha}_1$. This distribution can be used to examine the estimator's behavior and to assess the effects of simultaneity bias. Since the regressor y_2 is endogenous, the sampling distribution is expected to be centered away from the true parameter value α_1 , illustrating the inconsistency of OLS in simultaneous equations models.

```

8  %% 5. Sampling distribution of the OLS estimator
9
10 % 5.1. Set the number of simulations
11 N_sim = 1000;
12
13 % 5.2. Preallocate a vector for the simulated estimates of alpha_1
14 alpha_1_hat_sim = NaN(N_sim,1);
15
16 % 5.3. Create the sampling distribution
17 for i = 1:N_sim
18     z_1 = random('Normal',0,1,[N_obs 1]);
19     z_2 = random('Normal',0,1,[N_obs 1]);
20     u_1 = random('Normal',0,1,[N_obs 1]);
21     u_2 = random('Normal',0,1,[N_obs 1]);
22     y_1 = (beta_1 * z_1 + alpha_1 * beta_2 * z_2 + ...
23           u_1 + alpha_1 * u_2) / ...
24           (1 - alpha_1 * alpha_2);
25     y_2 = (alpha_2 * beta_1 * z_1 + beta_2 * z_2 + ...

```

```

26         alpha_2 * u_1 + u_2) / ...
27         (1 - alpha_1 * alpha_2);
28     X = [y_2 z_1];
29     LSS = lss(y_1,X);
30     alpha_1_hat_sim(i,1) = LSS.B_hat(1,1);
31 end

```

6. Bias of the OLS estimator

The bias of the OLS estimator is evaluated using the empirical sampling distribution obtained from the Monte Carlo simulation. First, the mean of the simulated estimates of $\hat{\alpha}_1$ is computed.

The bias is then calculated as the difference between the average estimated value and the true parameter value, α_1 . If OLS were unbiased, this difference would be close to zero. However, because y_2 is endogenous and correlated with the structural disturbance u_1 , the estimated bias is expected to be nonzero, providing further evidence of simultaneity bias in the first structural equation.

```

8 %% 6. Bias of the OLS estimator
9
10 % 6.1. Compute the mean of the sampling distribution
11 alpha_1_hat_mean = mean(alpha_1_hat_sim);
12
13 % 6.2. Compute the bias
14 bias = alpha_1_hat_mean - alpha_1;

```

7. Plot the sampling distribution of the OLS estimator

The empirical sampling distribution of $\hat{\alpha}_1$ is visualized using a kernel density estimate based on the Monte Carlo replications. The resulting density provides a graphical representation of the variability of the OLS estimator across repeated samples.

To assess the presence of simultaneity bias, the mean of the sampling distribution is plotted alongside the true parameter value, α_1 . If OLS were unbiased, the center of the sampling distribution would coincide with the true parameter value. Any systematic deviation between the two indicates bias arising from the endogeneity of y_2 . The figure therefore provides a visual illustration of the inconsistency of the OLS estimator in the simultaneous equations model.

```

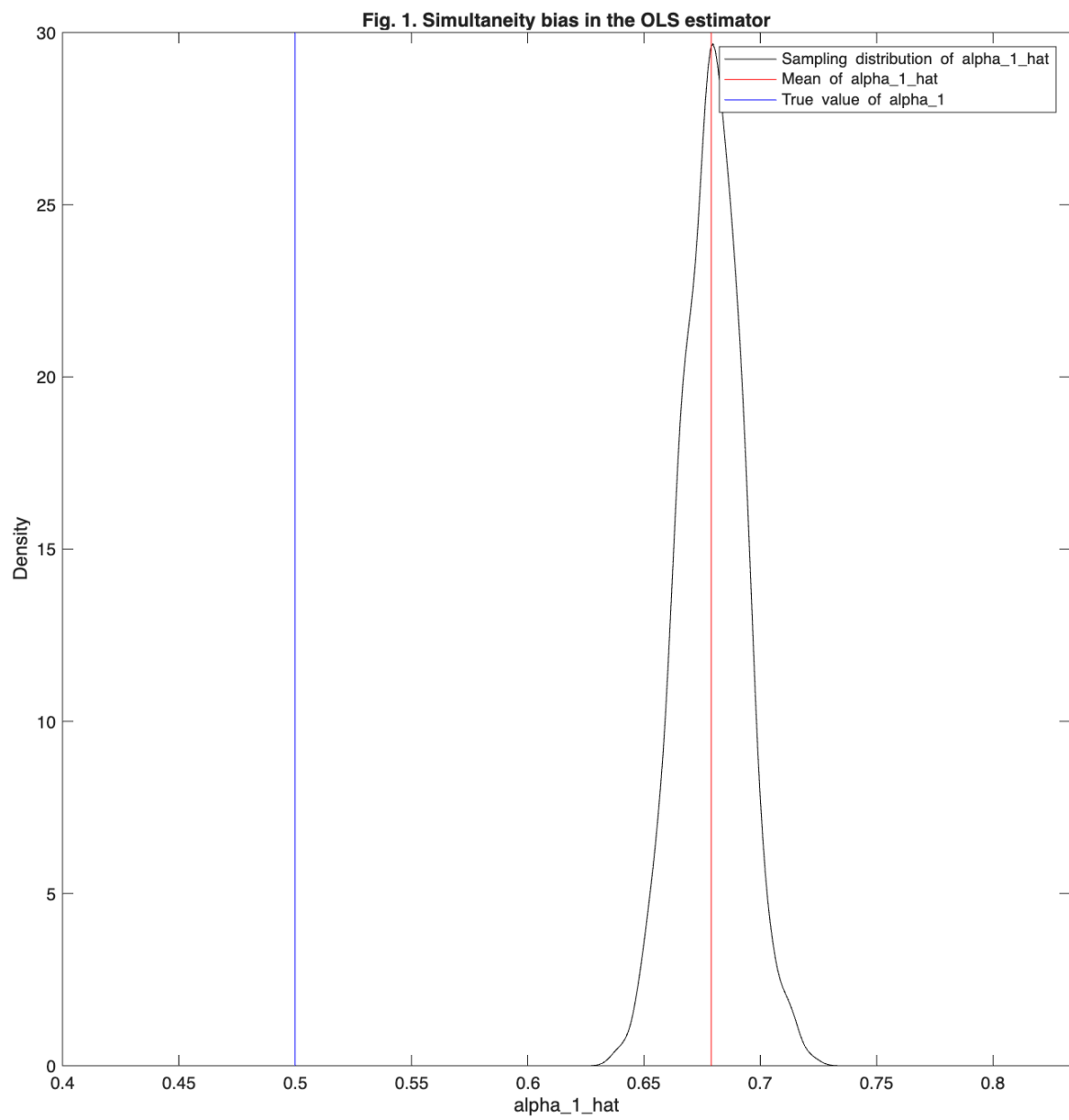
8 %% 7. Plot the sampling distribution of the OLS estimator
9
10 % Create a figure
11 figure;
12 % Set the figure size
13 set(gcf,'Position',[100 100 1000 1000]);
14 % Estimate the density
15 [f,x] = ksdensity(alpha_1_hat_sim);
16 % Plot the density
17 plot(x,f, ...
18      'Color',[0.000 0.000 0.000])
19 % Expand the x-axis range
20 xlim([alpha_1 - 0.1 max(x) + 0.1])

```

```

21 % Add the mean of the sampling distribution
22 hold on
23 line([alpha_1_hat_mean alpha_1_hat_mean],ylim, ...
24      'Color',[1.000 0.000 0.000])
25 % Add the true value
26 line([alpha_1 alpha_1],ylim, ...
27      'Color',[0.000 0.000 1.000])
28 % Add labels
29 title('Fig. 1. Simultaneity bias in the OLS estimator')
30 legend('Sampling distribution of alpha\_1\_hat', ...
31      'Mean of alpha\_1\_hat', ...
32      'True value of alpha\_1')
33 ylabel('Density')
34 xlabel('alpha\_1\_hat')
35 hold off

```



8. When does simultaneity bias disappear?

To illustrate the source of simultaneity bias, the coefficient α_2 is set equal to zero and the simulation exercise is repeated. In this case, the second endogenous variable no longer depends on the error term u_1 , eliminating the correlation between y_2 and the disturbance in the first structural equation. As a result, the exogeneity condition is restored and the OLS estimator becomes unbiased. The sampling distribution of $\hat{\alpha}_1$ is therefore expected to be centered much closer to the true parameter value α_1 , demonstrating that simultaneity bias arises from the feedback relationship that induces correlation between the endogenous regressor and the structural error term.

9 . Final notes

This file is prepared by Tunga Kantarcı. This file and the accompanying MATLAB files are available on GitHub and can be accessed via this [link](#).